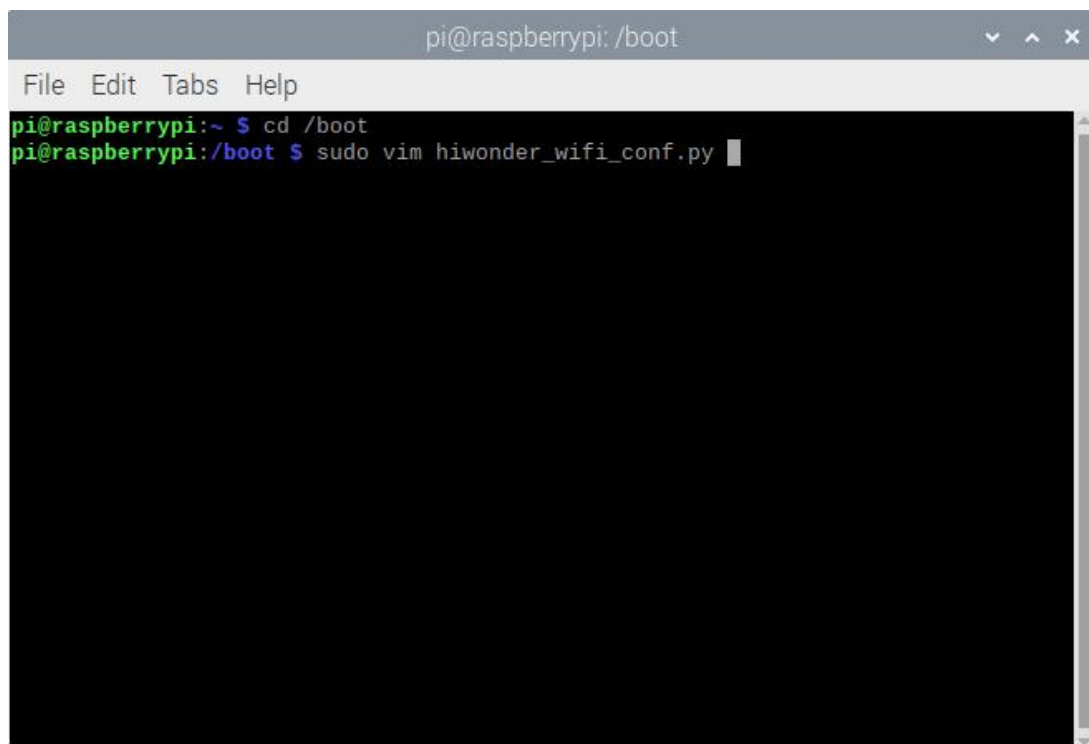


Lesson 6 Modify Wi-Fi

1. Modify Raspberry Pi Wi-Fi

The computer is supposed to connect to the Wi-Fi hotspot named with the first letters “HW” which is launched by Raspberry Pi. When there are multiple robots around, the wrong connection may occur. If you want to modify the default Wi-Fi name and password, please check the following steps.

- 1) Turn on Raspberry Pi, start VNC and then connect it to the Raspberry Pi remote desktop.
- 2) Press “Ctrl+Alt+T” to turn on LX terminal.
- 3) Enter the following command to jump and open the Wi-Fi configuration file with an editor.



```
pi@raspberrypi: /boot
File Edit Tabs Help
pi@raspberrypi:~ $ cd /boot
pi@raspberrypi:/boot $ sudo vim hiwonder_wifi_conf.py
```

4) Interface as below:

```
File Edit Tabs Help
1 #!/usr/bin/python3
2 #coding:utf8
3
4 #HM_WIFI_MODE = 2                #wifi working mode, 1 is AP mode, 2 is STA mode
5 #HM_WIFI_AP_SSID = 'ssid_name'    #SSID in AP mode,character and digital composition
6 #HM_WIFI_AP_PASSWORD = 'passwords' #WiFi password in AP mode, character and digital composition
7 #HM_WIFI_AP_GATEWAY = '192.168.149.1' #The local IP in AP mode, the default is 192.168.149.1, if you modify this item, you
# will not be able to enter the WiFi configuration interface on the mobile APP
8 #HM_WIFI_FREQ_BAND = 5            #the WiFi frequency in AP mode is directly assigned as 2.4 or 5 corresponding to 2.4G and 5G
9 #HM_WIFI_CHANNEL = 161            #WiFi signal channel in AP mode, under 5G testing available channels are 149, 153, 157
# 161
10 #HM_WIFI_STA_SSID = 'ssid_name'   #SSID in STA mode
11 #HM_WIFI_STA_PASSWORD = 'passwords' #WiFi password in STA mode
12 #HM_WIFI_TIMEOUT = 30              #the timeout when the STA connects to the Wi-Fi hotspot, if out time is not successful
# connection then the connection is considered as a failure. The default is 30 seconds
13 #HM_WIFI_LED = True                #if use LED indicator, the default is True, then use LED indicator
14 #HM_WIFI_RESET_NOW = False         # clear all configuration files, the default is False, when set to True, the program will
# clear all configurations and restore the initial state, including mobile phone configuration and manual editing of configuration files.
```

5) Press the "i" on the keyboard and then the "--Insert--" mark will be displayed in the lower left corner of the interface. Please refer to the corresponding notes to modify.

```
File Edit Tabs Help
1 !/usr/bin/python3
2 #coding:utf8
3
4 #HW_WIFI_MODE = 2                #wifi working mode, 1 is AP mode, 2 is STA mode
5 #HW_WIFI_AP_SSID = 'ssid_name'   #SSID in AP mode,character and digital composition
6 #HW_WIFI_AP_PASSWORD = 'passwords' #WIFI password in AP mode, character and digital composition
7 #HW_WIFI_AP_GATEWAY = '192.168.149.1' #The local IP in AP mode, the default is 192.168.149.1, if you modify this item, you
  will not be able to enter the WiFi configuration interface on the mobile APP
8 #HW_WIFI_FREQ_BAND = 5          #the WiFi frequency in AP mode is directly assigned as 2.4 or 5 corresponding to 2.4G and
  5G
9 #HW_WIFI_CHANNEL = 161          #WiFi signal channel in AP mode, under 5G testing available channels are 149, 153, 157
  , 161
10 #HW_WIFI_STA_SSID = 'ssid_name' #SSID in STA mode
11 #HW_WIFI_STA_PASSWORD = 'passwords' #WIFI password in STA mode
12 #HW_WIFI_TIMEOUT = 30           #the timeout when the STA connects to the Wi-Fi hotspot.if out time is not successful
  connection then the connection is considered as a failure. The default is 30 seconds
13 #HW_WIFI_LED = True             #if use LED indicator, the default is True, then use LED indicator
14 #HW_WIFI_RESET_NOW = False      # clear all configuration files, the default is False, when set to True, the program w
  ill clear all configurations and restore the initial state, including mobile phone configuration and manual editing of con
  figuration files.
```

6) If you want to modify the Raspberry Pi name as “Hiwonder” and password as “12345678”, only need to revise the info as shown in the below figure. Do not forget to delete “#” to make it effective.

```
1 #!/usr/bin/python3
2 #coding:utf8
3
4 #HW_WIFI_MODE = 2          #wifi working mode, 1 is AP mode, 2 is STA mode
5 HW_WIFI_AP_SSID = 'Hiwonder' #SSID in AP mode.character and digital composition
6 HW_WIFI_AP_PASSWORD = '12345678' #WIFI password in AP mode, character and digital composition
7 #HW_WIFI_AP_GATEWAY = '192.168.149.1'#The local IP in AP mode, the default is 192.168.149.1, if you modify this item, you
  will not be able to enter the WiFi configuration interface on the mobile APP
8 #HW_WIFI_FREQ_BAND = 5     #the WiFi frequency in AP mode is directly assigned as 2.4 or 5 corresponding to 2.4G and
  5G
9 #HW_WIFI_CHANNEL = 161     #WiFi signal channel in AP mode, under 5G testing available channels are 149, 153, 157
  , 161
10 #HW_WIFI_STA_SSID = 'ssid_name' #SSID in STA mode
11 #HW_WIFI_STA_PASSWORD = 'passwords'#WIFI password in STA mode
12 #HW_WIFI_TIMEOUT = 30       #the timeout when the STA connects to the Wi-Fi hotspot.if out time is not successful
  connection then the connection is considered as a failure. The default is 30 seconds
13 #HW_WIFI_LED = True        #if use LED indicator, the default is True, then use LED indicator
14 #HW_WIFI_RESET_NOW = False  # clear all configuration files, the default is False, when set to True, the program w
  ill clear all configurations and restore the initial state, including mobile phone configuration and manual editing of con
  figuration files.
```

7) If the wireless network card doesn't support 5G WiFi, please change the default value of HW_WIFI_FREQ_BAND from 5 to 2.4, so that the WiFi type is 2.4G. However, the 2.4G transmission rate will be lower than the 5G rate.

```
1 #!/usr/bin/python3
2 #coding:utf8
3
4 #HW_WIFI_MODE = 2          #wifi working mode, 1 is AP mode, 2 is STA mode
5 HW_WIFI_AP_SSID = 'Hiwonder' #SSID in AP mode.character and digital composition
6 HW_WIFI_AP_PASSWORD = '12345678' #WIFI password in AP mode, character and digital composition
7 #HW_WIFI_AP_GATEWAY = '192.168.149.1'#The local IP in AP mode, the default is 192.168.149.1, if you modify this item, you
  will not be able to enter the WiFi configuration interface on the mobile APP
8 HW_WIFI_FREQ_BAND = 2.4    #the WiFi frequency in AP mode is directly assigned as 2.4 or 5 corresponding to 2.4G and
  5G
9 #HW_WIFI_CHANNEL = 161     #WiFi signal channel in AP mode, under 5G testing available channels are 149, 153, 157
  , 161
10 #HW_WIFI_STA_SSID = 'ssid_name' #SSID in STA mode
11 #HW_WIFI_STA_PASSWORD = 'passwords'#WIFI password in STA mode
12 #HW_WIFI_TIMEOUT = 30       #the timeout when the STA connects to the Wi-Fi hotspot.if out time is not successful
  connection then the connection is considered as a failure. The default is 30 seconds
13 #HW_WIFI_LED = True        #if use LED indicator, the default is True, then use LED indicator
14 #HW_WIFI_RESET_NOW = False  # clear all configuration files, the default is False, when set to True, the program w
  ill clear all configurations and restore the initial state, including mobile phone configuration and manual editing of con
  figuration files.
```

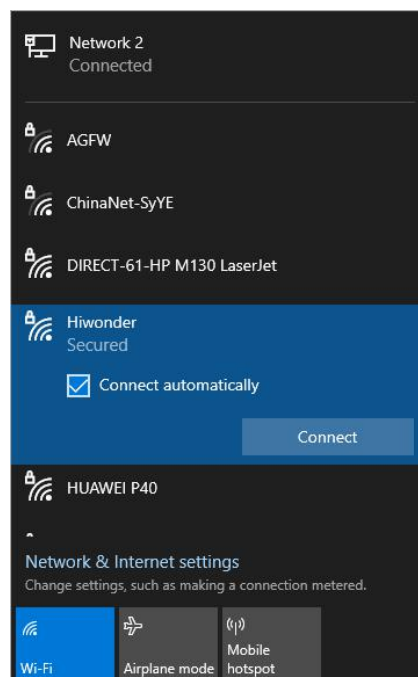
8) After modification, press “ESC” and enter “:wq”. Then save and exit the file.

```
1 #!/usr/bin/python3
2 #coding:utf8
3
4 #HW_WIFI_MODE = 2          #wifi working mode, 1 is AP mode, 2 is STA mode
5 HW_WIFI_AP_SSID = 'Hiwonder' #SSID in AP mode.character and digital composition
6 HW_WIFI_AP_PASSWORD = '12345678' #WIFI password in AP mode, character and digital composition
7 #HW_WIFI_AP_GATEWAY = '192.168.149.1'#The local IP in AP mode, the default is 192.168.149.1, if you modify this item, you
  will not be able to enter the WiFi configuration interface on the mobile APP
8 HW_WIFI_FREQ_BAND = 2.4    #the WiFi frequency in AP mode is directly assigned as 2.4 or 5 corresponding to 2.4G and
  5G
9 #HW_WIFI_CHANNEL = 161     #WiFi signal channel in AP mode, under 5G testing available channels are 149, 153, 157
  , 161
10 #HW_WIFI_STA_SSID = 'ssid_name' #SSID in STA mode
11 #HW_WIFI_STA_PASSWORD = 'passwords'#WIFI password in STA mode
12 #HW_WIFI_TIMEOUT = 30       #the timeout when the STA connects to the Wi-Fi hotspot.if out time is not successful
  connection then the connection is considered as a failure. The default is 30 seconds
13 #HW_WIFI_LED = True        #if use LED indicator, the default is True, then use LED indicator
14 #HW_WIFI_RESET_NOW = False  # clear all configuration files, the default is False, when set to True, the program w
  ill clear all configurations and restore the initial state, including mobile phone configuration and manual editing of con
  figuration files.
```

9) Enter “sudo systemctl restart hw-wifi.service” in the command line, press “Enter ” to restart the file. Then the VNC will disconnect automatically.

```
pi@raspberrypi:~ $ cd /boot
pi@raspberrypi:/boot $ sudo vim hiwonder_wifi_conf.py
pi@raspberrypi:/boot $ sudo systemctl restart hw-wifi.service
```

10) In the Wi-Fi setting area, you can find that the Wi-Fi name will be updated as “Hiwonder”. Enter password “12345678” to connect.



2. Set a Static IP for the Raspberry Pi

When using wired connection, it's more convenient for you to set a static IP. Here is the method of setting a static IP.

- 1) Turn on Raspberry Pi, start VNC and connect to Raspberry Pi remote desktop.
- 2) Press “Ctrl+Alt+T” to open LX terminal.
- 3) Enter the following command to open the dhcpd.conf configuration file with an editor.

```
pi@raspberrypi:~ $ sudo nano /etc/dhcpd.conf
```

```

pi@raspberrypi: ~
File Edit Tabs Help
GNU nano 3.2 /etc/dhcpd.conf

# A sample configuration for dhcpd.
# See dhcpd.conf(5) for details.

# Allow users of this group to interact with dhcpd via the control socket.
#controlgroup wheel

# Inform the DHCP server of our hostname for DDNS.
hostname

# Use the hardware address of the interface for the Client ID.
clientid
# or
# Use the same DUID + IAID as set in DHCPv6 for DHCPv4 ClientID as per RFC4361.
# Some non-RFC compliant DHCP servers do not reply with this set.
# In this case, comment out duid and enable clientid above.
#duid

# Persist interface configuration when dhcpd exits.
persistent

[ Read 65 lines ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line

```

4) Insert the following content at the end of the file.

```

static ip_address=192.168.11.51/24

static ip6_address=fd51:42f8:caae:d92e::ff/64

static routers=192.168.11.1

static domain_name_servers=192.168.0.1 8.8.8.8 fd51:42f8:caae:d92e::1

```

```

pi@raspberrypi: ~
File Edit Tabs Help
GNU nano 3.2 /etc/dhcpd.conf

# define static profile
#profile static_eth0
#static ip_address=192.168.1.23/24
#static routers=192.168.1.1
#static domain_name_servers=192.168.1.1

# fallback to static profile on eth0
#interface eth0
#fallback static_eth0

static ip_address=192.168.11.51/24
static ip6_address=fd51:42f8:caae:d92e::ff/64
static routers=192.168.11.1
static domain_name_servers=192.168.0.1 8.8.8.8 fd51:42f8:caae:d92e::1

```


Press “**Ctrl+o**” tp save the file and “**Ctrl+x**” to exit it.

5) Last step is enter “**sudo reboot**” command to restart Raspberry Pi and connect with the new static IP.